

Neil Mascarenhas

U.S. Citizen
New York, NY

mmm223@cornell.edu

Cell: 248.464.1432

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering in Financial Engineering, **GPA: 4.08**

Expected Dec. 2026

University of Michigan, Ann Arbor, MI
Bachelor of Science in Mathematics and Data Science, BBA in Finance, **GPA: 3.63**

Dec. 2024

Selected Coursework: Stochastic Calculus, Bayesian Estimation, Robust and Stochastic Optimization, Discrete State Stochastic Processes, Numerical Methods, Applied Functional Analysis, Simulation Modeling and Analysis

SKILLS

Technical: Python (quant research, market simulations, ML, PyTorch), R (Bayesian statistics, MCMC), C++, SQL, Ollama

EXPERIENCE

Quantitative Analyst Intern, *BHG Financial*, Syracuse, NY **Jun. to Sept. 2026**

- Built LLM-forward selection pipeline for segmenting over 20 million customers, adding data governance, standardization, and auditability to a previously opaque manual selection process, with an 80% efficiency increase.
- Engineered LLM tools, structured workflows, and validation guardrails to constrain model behavior and verify intermediate outputs, iteratively improving reliability and alignment with intended selection logic.
- Modeled underwriter approval process to calculate probability of overturning the automated decision, allowing the ability to evaluate individual underwriter approval percentages against a standardized benchmark.

Research Assistant, *University of Michigan*, Ann Arbor, MI **Jan. to Aug. 2025**

- Engineered feed-forward neural networks in Python to model exploration dynamics in N-player market simulations, extending bounded logit-demand models to unbounded environments.
- Conducted large-scale simulations analyzing convergence of collusive equilibria; validated theoretical models through empirical experimentation and algorithmic testing.
- Implemented exploration mechanism variants and evaluated convergence sensitivity across neural network designs.

Quantitative Analyst Intern, *Yousif Capital Management*, Bloomfield Hills, MI **May to Aug. 2022**

- Applied Isolation Forest and statistical filters to 1.5M trade records, detecting anomalous risk exposure patterns.
- Built alternative data pipeline processing 5,000+ keywords to extract/rank market intelligence from external datasets.

Quantitative Developer Intern, *Greenstone Farm Credit Services*, East Lansing, MI **May to Dec. 2021**

- Designed Selenium automation to process 5,000+ client tax forms, cutting 4,800 labor hours and saving \$30K annually.
- Developed R fuzzy-matching algorithms for address parsing and entity linkage, achieving 100% mapping coverage.

LEADERSHIP EXPERIENCE & PROJECTS

President | Founder, *Quantitative and Computational Finance Group*, Ann Arbor, MI **Sept. 2021 to Aug. 2025**

- Founded and scaled quantitative finance group to 50+ members; led projects spanning options pricing, market microstructure, systematic strategies, statistical arbitrage, portfolio optimization, and NLP.
- Coded C++ SABR options pricing engine and Monte Carlo calibration for detection of derivative mispricings.
- Designed lead-lag equity trading strategies using network theory; optimized backtests with Python numba.

Heavy Tailed Risk Modeling, *Cornell University*, Ithaca, NY **Oct. 2025**

- Simulated ruin and maximum drawdown probabilities under heavy-tailed return processes using Monte Carlo/IS.
- Derived Gaussian scale-mixture representation to accelerate simulation and analyzed sensitivity of drawdown probability to tail index alpha between 1.4 and 1.7.

ACTIVITIES/INTERESTS & CERTIFICATES

Heavy tails, alpha-stable processes, LLM output parsing, automation, yoga, cooking, meditation, reading, hiking, traveling
Certificates: Baruch Pre-MFE Numerical Linear Algebra for Financial Engineering (Distinction), Bloomberg Market Concepts, Akuna Capital Options 101